



Name
Class
R. No.



Instructions

1. The test contains six questions

Q1: A man wishes to estimate the distance of a nearby tower from him. He stands at a point A in front of the tower C and spots a very distant object O in line with AC. He then walks perpendicular to AC up to B, a distance of 100 m, and looks at O and C again. Since O is very distant, the direction BO is practically the same as AO; but he finds the line of sight of C shifted from the original line of sight by an angle $\theta = 40^\circ$ estimate the distance of the tower C from his original position A.

10 Marks

Q2: The moon is observed from two diametrically opposite points A and B on Earth. The angle θ subtended at the moon by the two directions of observation is $1^\circ 54'$. Given the diameter of the Earth to be about 1.276×10^7 m, compute the distance of the moon from the Earth.

10 Marks

Q3 Give a simple method for estimating the molecular size of oleic acid.

10 Marks

Q4: Distinguish between two terms: **accuracy** and **precision**. Elaborate with the help of examples. 10 Marks

Q5: Two clocks are being tested against a standard clock located in a national laboratory. At 12:00:00 noon by the standard clock, the readings of the two clocks are :

	Clock 1	Clock 2
Monday	12:00:05	10:15:06
Tuesday	12:01:15	10:14:59
Wednesday	11:59:08	10:15:18
Thursday	12:01:50	10:15:07
Friday	11:59:15	10:14:53
Saturday	12:01:30	10:15:24
Sunday	12:01:19	10:15:11

If you are doing an experiment that requires precision time interval measurements, which of the two clocks will you prefer ?

10 Marks

Q6 Find the relative error in Z, if $Z = \frac{A^2 + B^{\frac{3}{2}}}{CD^{\frac{1}{5}}}$

20 Marks